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The varying effect of eWOM valence on online game consumption: An empirical examination focusing on the user characteristics of social media

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Abstract

This paper examines the effects of electronic word-of-mouth (eWOM) on game consumption, with a focus on the Korean online PC game industry. We analyze how the volume of positive and negative reviews affects consumers differently, depending on the characteristics of the platform on which they are posted. We categorize social media platforms according to their degree of homophily and investigate the differences between high homophily and low homophily platforms in terms of the impact of eWOM valence. The results indicate that positive eWOM in both types leads to increased consumption, whereas negative eWOM has varying effects based on the level of homophily on the social media platforms. Negative eWOM from high homophily social networks decreases game consumption, while those from low homophily social networks increase it. This study also finds that eWOM has a greater impact on socially interactive games. We employ Panel Vector Autoregressive (PVAR) modeling and separate estimation by genre and find robust results. The study highlights the importance of applying marketing strategies considering the characteristics of each platform.

Keywords: eWOM, social media platforms, online PC game, panel vector auto-regression (PVAR)

1. Introduction

Increasing consumer access to information and opportunities for sharing product information, such as electronic word-of-mouth (eWOM), have been made possible by the development of the Internet [32]. It has previously been found that eWOM is related to consumer selection of products or brand image [46]. Consumer reviews and search trend data on major search engines have a significant correlation. The characteristics of eWOM have been shown to influence consumer behavior [30].

Many studies have examined how the volume and valence of eWOM affect consumers [5, 13, 20]. Researchers have confirmed that eWOM volume and positive eWOM are beneficial to firms; however, the magnitude and direction of negative eWOM have remained unclear [20]. Some studies have also suggested that negative eWOM can convey favorable effects as it draws attention [6, 49]. The discrepancies in the literature can be attributed to the diverse characteristics of social media. Previous research has shown that each social media platform has different types of user engagement, resulting in varying characteristics and user-generated content. Accordingly, reviews with similar tone and content are likely to have different effects on consumers based on the platform on which they are posted.

Despite this, few studies have examined the characteristics of multiple social platforms and have examined the relationship between these platforms and consumer behavior [18, 28]. The aim of our research is to fill this gap by utilizing data from multiple social media platforms to examine how the volume of positive and negative reviews affects consumers differently, depending on the characteristics of the platform on which they are posted. Based on Kitchens et al. [41] method, we categorize social media platforms according to their degree of homophily. In this study, we investigate the differences between the two groups based on the impact of eWOM valence. We aim to gain a comprehensive understanding of the effects of eWOM that arise from the unique characteristics of each social media platform.

The purpose of our research is to explore the relationship between eWOMs and online game consumption. We investigate this by analyzing online game consumption data from the Korean online PC game industry. Our dataset includes daily observations from January 17, 2019, to January 17, 2020, which cover 168 game brands and the sum of the daily hours played by each game brand. We also collected text data from six social media platforms in Korea, including Facebook, Twitter, Instagram, YouTube, Communities, and News sites. We employ Panel Vector Autoregressive (PVAR) modeling to illustrate the dynamic effect of the different types of social media. Our results indicate that positive eWOM in both types (i.e., high homophily vs. low homophily) leads to increased consumption, whereas negative eWOM has varying effects based on the level of homophily on social media platforms. High-homophily social networks decrease game consumption, while low-homophily social networks increase it. Also, a subgroup analysis showed that eWOM has a greater impact on socially interactive games. We find robust results even when genres are separately estimated.

We contribute to the social media literature by providing empirical evidence of the effect of eWOM according to the existing social media framework. The richness in our data and method allows us to divide social media according to their characteristics and directly link the effect it brings to consumer behavior. Our research also has several managerial contributions. We give implications to managers on how to utilize social media with high homophily. Our research also suggests that analyzing the characteristics of social media helps to come up with strategies as it provides meaningful guidance to the direction and the persistence of social buzz.

The remainder of this paper is organized as follows. We first examine the relevant literature about eWOM, social media, and its effect on consumption. Then, we introduce our model and data used in this study, followed by results, discussion, and conclusion.

2. Literature Review and Theoretical Framework

2.1 eWOM: The Influence of Social Media

Word-of-mouth (WOM) can be a powerful tool for influencing consumer decisions since it provides greater credibility than traditional forms of advertising [12]. Customers perceive WOM differently than formal advertisements, where an ad is seen as an instrument for selling products, while customers view WOM as an interpersonal exchange with goals beyond mere persuasion [8]. Moreover, WOM provides confirmation and social support, thus assuring participants that they have made the best choice when purchasing a product [29, 50].

The rise of the internet and social media has given unprecedented power to WOM, leading to the development of electronic word-of-mouth (eWOM). Consumers are able to share information quickly and with a wide audience by using online platforms [12, 54]. As eWOM messages are posted on online platforms, they tend to persist longer than traditional face-to-face conversations [12, 32]. It has become increasingly imperative for companies to monitor and manage social chats because consumers research products and services prior to making a purchase [47, 59].

Past research has shown that eWOM plays a critical role in shaping consumer behavior for experience goods (e.g., movies, games, tourism services, etc.) [52, 61]. When purchasing experience goods, consumers may find it difficult to assess the quality of the product before purchasing, and therefore, turn to online communication to reduce uncertainty [52]. Online comments on movies, for example, can significantly influence a movie's daily box office performance, providing non-commercial opinions from audiences and some professional opinions from critics [5, 20, 22, 26, 43]. Similarly, hotel reviews posted on social media and travel guide websites can significantly impact the choices of other tourists [54, 60, 65].

In a similar vein, in the video game industry, social interactions among gamers on the internet have been one of the major driving forces of its growth. Gaming companies use a variety of social media channels to promote their products and attract new consumers. Game users interact with other users and exchange game information via social media channels [4]. Especially, online gaming communities have strong ties to social media, which creates a network effect. Therefore, it is crucial

for game publishers to engage individual players in a general discussion of the content to influence potential users.

2.2 Social Media Classification

The term "social media" refers to a collection of internet-based applications that facilitate generating, exchanging, and sharing of user-generated content [36]. Social media platforms offer unprecedented opportunities for users to express their reactions [1]. Furthermore, social media provides an opportunity for consumers to voice their opinions about products and services, both positively and negatively.

As social media platforms have different types of user-generated content, we aim to group the platforms according to their tendencies and examine the impact of each group's eWOM on game consumption. To categorize social media platforms, we refer to Kitchens et al. [41]. They provided a framework for how social buzz is exposed and consumed based on social media platforms' characteristics. Two key factors determining social buzz exposure are social network homophily and algorithmic filtering.

Homophily refers to the phenomenon in which communication occurs more frequently between individuals with similar interests than between individuals of different backgrounds. There are social media platforms where users with similar contexts have a higher tendency to connect constantly, resulting in a homogenous social network. Accordingly, homophily affects the information that users receive and share on social media, as well as the communication between like-minded individuals [45].

Another important characteristic for grouping social media platforms is the extent of algorithmic filtering. Based on prior research, social media platforms prioritize user-generated content differently depending on how much algorithm filtering is involved in the prioritization process [41]. Algorithms embedded in social media platforms play an increasingly important role in sorting information based on relevancy [27]. By applying users' personal information to algorithms, personalized user

experiences can be created and are an important feature for social media platforms aiming to build user loyalty [51]. In recent years, a number of social media platforms that did not previously offer algorithmic filtering are now implementing algorithms for the benefit of their users.

(Insert Table 1 about here.)

(Insert Table 2 about here.)

2.3 eWOM Valence in the Game Industry

Previous research has shown that users perceive and interact differently based on the level of homophily on social media platforms. Social media platforms with high homophily tend to form groups of individuals who share similar views, and algorithms aim to maximize user satisfaction by displaying content that aligns with their interests [38]. As such, users are often restricted to a limited range of information and confined to small information bubbles or echo chambers [9, 11, 48]. For example, Facebook and Twitter are more affected by the echo chamber effect and are dominated by homophilic clusters [15]. Thus, echo chambers can result in biased views and uncritical acceptance of information, including the spread of fake news [55].

Suwandee [56] argue that negative eWOMs provided by homophilic groups significantly change attitudes. For example, in high homophily groups, there was a high level of consensus on issues, and members tended to accept the majority opinion, resulting in a change of attitude [56]. In addition, homophily increases the adoption of new behaviors among community members [10]. It appears that the number of friends who have already adopted a behavior (i.e., consensus) increases the probability of the individual adopting it [37].

Sharma & Jain [53] found that homophily and social cognitive theory factors, including social outcome expectations and self-efficacy, significantly affect consumers' perception of power and control on social media. People tend to believe information that aligns with their pre-existing views regardless of reliability due to confirmation bias [39]. The posted article is more likely to trigger both emotional and cognitive responses when it is more relevant to the preexisting opinion that the user has

[40]. Therefore, in social media with high homophily, users are more likely to agree with opinions, whether positive or negative.

In contrast, when users are searching for valuable information, they prefer social media with low homophily. They use these channels to obtain useful information about products or services rather than to establish and maintain personal relationships. For example, social news sites that provide recreational value or enlightenment are highly preferred by users [7]. As a result, consumers who read reviews on these platforms tend to focus on the information provided regarding products rather than being influenced by the sentiments of others – for example, Yang et al. [63] analyzed a review dataset in online forums and found that readers were primarily interested in product information rather than forum involvement.

Past research mentioned some benefits of negative eWOM. They are more likely to grab people's attention, and users often feel that the negative eWOMs are the most helpful in making purchase decisions [42, 60]. Especially for experience goods, additional images and detailed objective descriptions in online negative reviews could encourage consumers to make purchasing decisions [64]. In this sense, it could be said that negative eWOM raises brand awareness by grabbing people's attention. It also helps consumers make better buying decisions and facilitates user experience as it provides potential buyers with helpful information that cannot be realized without other people's experiences. Thus, along with the positive eWOM, we expect that negative eWOM in these social media platforms will also positively affect consumer's use of experience goods.

Our research builds upon the existing literature by examining how eWOMs in social media with high and low homophily affect consumer gaming consumption differently. Therefore, the objective of this study is to investigate the dynamic effect of different types of social networks to determine whether the results are consistent with these two groups' known characteristics.

(Insert Figure 1 about here.)

3. Method

3.1 Data

Social factors are especially important in games since potential users are more likely to consume games if they have more information about them. Through in-game communication and group functions, recent games have integrated social functions into gameplay. For instance, in 2015, Blizzard enabled linking Twitter accounts with World of Warcraft game accounts, which enabled in-game screenshots to be automatically posted to the linked Twitter account [4]. Accordingly, this study examines how different types of social media affect game consumption.

To empirically test our research questions, we used a multi-source dataset to understand how social buzz across different channels affects online PC game performance in South Korea. We collected user gameplay history data between January 17, 2019, and January 17, 2020, including daily user counts, total hours played, and game characteristics. In this period, 342 games were available on the Korean online PC gaming market. To reduce aggregation bias, we selected 168 game brands that were available daily during the study period, with the total observations of 61,320.

We collected daily social media buzz of 168 selected game brands across six platforms: Facebook, Twitter, Instagram, YouTube, Communities, and News sites. We defined eWOM volume as the number of times information has been delivered electronically and treated the positive and negative volumes of eWOMs in the same manner. The text data were preprocessed and divided into three categories based on their valence: positive, negative, and neutral. As part of the preprocessing, we used AI engines such as CNN (Convolutional Neural Networks), RNN (Recurrent Neural Networks), Bi-LSTM (Bidirectional Long Short Term Memory), Naive Bayes, Logistic Regression, TF-IDF, and Word2Vec.

As previously noted, we classified social media metrics into two groups based on the platform's level of homophily. According to Bakshy et al. [3] and Vaidhyanathan [58], Facebook has a high level of homophily. Since Instagram is derived from Facebook, it exhibits a similar segment and is classified the same way. In addition, Twitter exhibits a homophilic tendency as well [41].

Social media channels with low homophily filter user posts based on their awareness of the topic, such as online communities (e.g., Reddit), online news sites, and YouTube [57]. For example, Reddit is classified as having a high level of algorithmic filtering and low homophily tendencies. Korean online communities exhibit similar characteristics in that users can communicate one-to-many within the platforms [35].

For news sites, an algorithm places the most popular articles at the top of the page, resulting in a high level of algorithmic filtering and a low level of homophily [57]. A YouTube algorithm recommends videos to its users and allows them to freely access all content, regardless of topic [17], which indicates YouTube has low homophily and high algorithm filtering.

3.2 Descriptive Statistics

Table 3 presents descriptive statistics for our main variables. In the study period, gamers spent 49,264,013 seconds on each game. The total volume of positive eWOM in Q2 (high homophily & low algorithm filtering) had the largest value (18.3731), followed by negative eWOM of Q1 (high homophily & high algorithm filtering), positive eWOM of Q1, and negative eWOM of Q2. The panel unit root test indicates that all series are stationary, which means they are suitable for PVAR analysis.

(Insert Table 3 about here.)

3.3 Measures

We used the total playtime of game i on day t as an indicator of game consumption. We grouped eWOMs from Facebook, Instagram, and Twitter into Q1 and YouTube, Online Communities, and News Sites into Q2. To gain a comprehensive overview of our collected eWOMs, each source was further divided into positive and negative sentiments, resulting in a total of four variables. Each group's positive and negative eWOMs are referred to as 'Q1_PWOM,' 'Q1_NWOM,' 'Q2_PWOM,' and 'Q2_NWOM.' There were several control variables, including the number of days after the game was released (days_since_release). We included the weekend variable to capture the time when consumers were most likely to play games. A vacation variable was also included to consider when

schools took summer and winter breaks. For instance, the vacation variable was set to 1 when schools were on summer or winter break and 0 when they were not.

3.4 Model Specification

For empirical analysis, we used a dynamic modeling framework using the Panel Vector Auto-Regression (PVAR) method. The PVAR enables more effective estimation of dynamic effects compared to conventional vector auto-regression (VAR) methods and allows the exploration of dynamic relationships between all variables [34, 44]. Using this model, we investigated how the volume of eWOMs from each group dynamically affected game performance over time.

To examine the interactions among social media channels and total play time, we conducted Granger causality tests to identify any potential endogeneity between pairs of our key variables [23]. The PVAR analysis was then conducted to understand the dynamic relationships between all variables, estimate and interpret regression coefficients, create and analyze impulse response functions, and calculate elasticities.

The main focus of our study was on how sentiments expressed by two distinct groups of eWOMs affect daily game consumption differently. *PlayTime* denotes the total amount of time game *i* is played in day *t* ($t = 1, 3, \dots, T$). Our main model comprises endogenous variables and their own preceding variables, which are the groups of eWOMs (Q1_PWOM, Q1_NWOM, Q2_PWOM, Q2_NWOM). Finally, we control for the *day_since_release*, *Weekend*, and *Vacation* in our analysis. Our PVAR model is specified as follows:

$$\begin{bmatrix} PlayTime_{i,t} \\ Q1_PWOM_{i,t} \\ Q1_NWOM_{i,t} \\ Q2_PWOM_{i,t} \\ Q2_NWOM_{i,t} \end{bmatrix} = \begin{bmatrix} \alpha_1^i \\ \alpha_2^i \\ \alpha_3^i \\ \alpha_4^i \\ \alpha_5^i \end{bmatrix} + \sum_{j=1}^J \begin{bmatrix} \beta_{1,1}^{ij} \dots \beta_{1,5}^{ij} \\ \beta_{2,1}^{ij} \dots \beta_{2,5}^{ij} \\ \beta_{3,1}^{ij} \dots \beta_{3,5}^{ij} \\ \beta_{4,1}^{ij} \dots \beta_{4,5}^{ij} \\ \beta_{5,1}^{ij} \dots \beta_{5,5}^{ij} \end{bmatrix} \begin{bmatrix} PlayTime_{i,t-j} \\ Q1_PWOM_{i,t-j} \\ Q1_NWOM_{i,t-j} \\ Q2_PWOM_{i,t-j} \\ Q2_NWOM_{i,t-j} \end{bmatrix} + \begin{bmatrix} \tau_{1,1}^{ij} \dots \tau_{1,3}^{ij} \\ \tau_{2,1}^{ij} \dots \tau_{2,3}^{ij} \\ \tau_{3,1}^{ij} \dots \tau_{3,3}^{ij} \end{bmatrix} \begin{bmatrix} day_since_release_{i,t} \\ Weekend_{i,t} \\ Vacation_{i,t} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t}^i \\ \varepsilon_{2,t}^i \\ \varepsilon_{3,t}^i \\ \varepsilon_{4,t}^i \\ \varepsilon_{5,t}^i \end{bmatrix}$$

Log transformation was used to address concerns about heteroscedasticity. To determine the appropriate lag length, the overall coefficient of determination (CD) was calculated, which measures

the proportion of variation explained by the PVAR model. We selected a one-lag model with the highest CD and Hansen's J. In the PVAR model, coefficients represent relationships between the lagged values of the variables (e.g., total gameplay and social buzz) and the left-hand side variables. Lastly, "GMM-style" instruments were employed to enhance model estimation.

4. Result

To test for unit roots in the data and estimate the stationary properties of each variable, we used the Levin-Lin-Chu (LLC) test, Harris-Tzavalis unit-root test, and ADF-Fisher test. All variables in the model were significant ($p < 0.001$), suggesting that they were stationary. Our findings allowed us to conduct a full dynamic system analysis through PVAR.

4.1 Estimation of the Model

The impulse-response functions (IRF) depict the prolonged shock that each variable gives to the total playtime presented in Table 4. We used 200 Monte Carlo replications and found that the shock's responses were significant and did not contain 0 when we plotted 95% confidence intervals.

(Insert Table 4 about here.)

The first column shows the coefficient of the estimated results in model (1), the playtime equation. All variables had a significant impact on total playtime. Moreover, Q1_PWOM (0.04, $p < 0.05$), Q2_PWOM (0.16, $p < 0.01$), and Q2_NWOM (0.10, $p < 0.01$) had a positive effect on total play time, while Q1_NWOM had a negative effect (-0.08, $p < 0.05$). An increase in eWOM volume from Q2 led to increased game consumption, regardless of its valence, while Q1_NWOM led to a significant decrease in game consumption. This implies that the valence of social buzz in high-homophily social media shapes consumers' attitudes toward certain products, whereas it is relatively less important for social media with low homophily.

4.2 Forecast Error Variance Decomposition (FEVD)

We used forecast error variance decomposition (FEVD) to determine the contribution of each eWOM volume group to total playtime. The FEVD illustrates to what extent changes in one dependent variable can be attributed as a result of shocks from the other time series variables. In addition to providing insights into their short-term effects, this also illustrates their long-term dynamics and variables' interdependency.

(Insert Table 5 about here.)

The explanatory power of each variable is shown in Table 5. Q2_PWOM had the highest explanatory power (average 0.499%), followed by Q1_PWOM (average 0.091%) and Q2_NWOM (average 0.046%). Q1_NWOM had the lowest explanatory power among the eWOMs (average 0.019%). Q2_PWOM also demonstrated the most significant explanatory power, indicating that it was the strongest predictor of playtime in the sample. In contrast, despite the significant negative effect of Q1_NWOM on playtime, the magnitude of this effect was relatively small across the entire sample. Comparing the explanatory power of each variable, we observed that the PWOM in both social media groups had a higher overall explanatory power than the NWOM.

4.3 Orthogonalized Impulse Response Function (OIRF)

(Insert Figure 2 about here.)

Next, we examined the impulse response function to examine the dynamic relationship between the variables and the playtime. It showed how the preceding variables affected $PlayTime_{i,t}$. Similar to the FEVD results, Q1_PWOM, Q2_PWOM, and Q2_NWOM exerted significantly positive effects on the dependent variables. The effect of $PlayTime_{i,t-1}$ gradually was reduced to zero, while the effect from all the other eWOM variables peaked after certain days, and then started to disappear.

The impulse response function (IRF) was then used to investigate how the variables affected $PlayTime_{i,t}$ by examining how the preceding variables affected $PlayTime_{i,t}$. Similar to the FEVD results, Q1_PWOM, Q2_PWOM, and Q2_NWOM contributed significantly to online game consumption. The effect of $PlayTime_{i,t-1}$ was gradually reduced to zero, while the effect of all other

eWOMs peaked after a certain number of days and then started to disappear. Despite Q2_PWOM having the greatest effect, Q1_PWOM persisted longer. This may be explained by the echo chamber effect that occurs more frequently on homophilic social media, resulting in longer-lived consequences [16].

4.4 Panel Impulse Response Functions (IRF): Subgroup Analyses

Specifically, we aimed to examine whether the results varied depending on game characteristics. To achieve this, we estimated PVAR models for different genres of games. Specifically, we estimated the model for seven different genres of games, including RPG (role-playing game), FPS (first-person shooter game), RTS (real-time strategy), Sports, Action, and Arcade. Additionally, we investigated whether the coefficients and explanatory power differed depending on the feature of the game. According to previous studies, eWOM volume and valence had a greater impact on game consumption for games containing social interaction features. We compared the results between games that provided a 'teampay' option and those not based on social features. A summary of the PVAR estimation results for each subgroup is presented in Table 6.

(Insert Table 6 about here.)

The columns (2) to (7) in the table show results when the samples are grouped by genre. Some coefficients had less significant *p*-values due to the smaller sample size, but among those that were significant, the results were consistent with the main model. This illustrates the robustness of the results since eWOM from each group exhibited similar patterns of effects, regardless of genre. The only exception was RTS, where Q2_NWOM negatively affected playtime, indicating that negative eWOM from Q2 decreased game consumption. The different results in RTS could potentially be attributed to the disproportionately large market share of RTS games in the Korean online game market; RTS games represent 46% of the total market share. In other words, most Korean users may have experienced and have sufficient knowledge of RTS games. Therefore, they may tend to focus more on their personal opinions and feedback rather than the information provided by NWOM. Columns (8) and (9) show differences in coefficients and significance between games played in a

team and those that are not. The results indicate that games with team-play functions were significantly affected by eWOMs, but non-team-play games were not. Table 7 illustrates the magnitude of the impact based on FEVD results.

(Insert Table 7 about here.)

There is a difference in explanatory power between the two groups. In games with a teamplay component, all types of eWOM had a greater explanatory power than those without a teamplay component. For teamplay games, Q2_PWOM contributed 0.043% of the variance of playtime, while for non-teamplay games, it contributed only 0.01%. Additionally, for social media with high homophily, positive comments had a much higher positive effect on teamplay games compared to non-teamplay games, where positive comments had almost no effect. While Q1_PWOM had the lowest explanatory power in the main analysis results, it had the second highest power with the teamplay games (average 0.133%). Comparing the coefficients of Q1_NWOM and Q2_NWOM, we also observed that the coefficients of teamplay games were about 0.04% higher than those of non-teamplay games. This proves that the consumption of socially interactive games depends more on the valence of the eWOM content, which is in line with the past findings [67].

5. Robustness Checks

In addition to our main model, we also utilized an alternative variable, UserCount, to provide another measure of online game usage. PlayTime shows the total time spent playing the game, while UserCount shows the number of players each day. The PVAR estimation results and subsample analyses are summarized in Tables 8 and 9.

(Insert Table 8 about here.)

(Insert Table 9 about here.)

Estimation for UserCount showed similar results with the previous model. The coefficients of Q1_PWOM, Q2_PWOM, and Q2_NWOM were positive and significant. The results indicate that

word of mouth was successful in attracting more users and attention to the games. Conversely, the results of Q1_NWOM indicate negative comments on social media with high homophily discouraged present or potential users from playing games. Similar results were obtained in a subsample analysis using UserCount as the dependent variable, indicating the number of game users is an appropriate measure for the robustness test.

6. Conclusion and Discussion

Our study examined the dynamic impact of eWOMs on online game consumption, and our extensive data allowed for a variety of analyses to determine how eWOMs influence online game consumption depending on their valence, origin, and product type. By using the PVAR method, we determined how total time consumption is affected by its own lags, as well as other eWOM variables, providing a dynamic depiction of how this effect varies over time. Thus, we were able to compare the persistence of the effect between the two groups, which depicted that the effect of highly homophilic social media is longer-lived. Users' tendency to adapt to group opinions and echo chambers contributed to longer shocks from the spread of eWOMs.

Overall, we observed that eWOMs in Q2 had a positive impact on game consumption regardless of their valence. In contrast, the valence in Q1 had a differential impact on game consumption. The results confirm our hypothesis that people follow the majority opinion in highly homophilic social media. The results further confirm our hypothesis regarding Q2, that people are less influenced by others' attitudes. In addition, exposure to certain products (herein games) led to increased consumption regardless of sentiment. Moreover, the sub-sample analysis indicates that socially interactive games are influenced by eWOMs more than non-interactive games. For robustness checks, we divided samples into genres and then used the number of users as the dependent variable.

6.1 Theoretical Contribution

First, we contribute to eWOM literature by providing empirical evidence for the social media framework. Previous literature commonly argued that the degree of homophily could be a way to

characterize different social media. However, few divided various social media into groups based on this theoretical basis and observed how this framework may provide the explanation for the influence of eWOM. Kitchens et al. [41] grouped social media based on the degree of homophily and denoted that the social media group with high homophily created echo chambers, but they did not provide empirical results that directly associated the different characteristics to user consumption. Our study used a PVAR model to depict the effect of eWOM on consumer behavior, which gives empirical support to the framework in Kitchens et al [41].

Second, we provide results by analyzing the effect of eWOM on multiple social media platforms. Previous literature that examined the effect of eWOM only examined one or two platforms. This resulted in a major limitation as social media platforms differ in their interaction among users due to the difference in algorithm, user characteristics, and interaction networks (e.g., reply, mention, follower, or retweet networks for Twitter). Our study considers the fact that these different characteristics of interactions bring significant differences in the results by incorporating the six most representative social media platforms into our research data. This multi-faceted approach brings an extension to the existing social media literature.

Third, the richness of our data provides robust results showing the causal relationship between the characteristics of social media and gaming behavior. Our data includes eWOM from social media, which have been neglected in previous research, such as Instagram, News sites, and online communities. We thereby found that the categorization based on homophily also holds in these social media. For the measurement of game consumption, we used panel data of 168 games, which constitute a major portion of the Korean online game industry.

Lastly, we use the methodological advantage of PVAR for our analysis. Since the volume of eWOM in each platform is related, using PVAR, which takes into account the endogenous effect of each variable, is preferable. Also, the PVAR method enabled us to depict how the magnitude of the effect varies over time. With this, we highlighted the longer persistence of the positive eWOM of Q1.

6.2 Practical Implications

This study has several practical implications. First, it highlights the importance of eWOM on homophilic social media platforms that have received less attention in the past. Prior research has primarily examined the relationship between eWOM on review sites, such as e-commerce websites, and product sales. Our findings indicate that social media platforms with high homophily, such as Facebook or Instagram, also generate significant eWOMs. As a result, consumers are not only influenced by the content of eWOMs but also by the user-friendliness of social networking sites. When formulating social media strategies, companies should take into account these differences in user characteristics. Second, game companies can effectively manage the social buzz surrounding their games based on our results. As positive eWOM related to games had a longer impact in Q1, companies should encourage consumers to leave positive comments on social media (e.g., Instagram, Facebook). Finally, companies can consider the persistence of impact within each eWOM group when developing strategies to respond to consumers. It has been shown that negative eWOMs generated from social media last longer than positive eWOMs. Therefore, companies must respond quickly to consumer in the wake of negative eWOMs.

6.3 Limitation and Further Research

Despite several useful insights, there are also some limitations, mainly due to the need for more data. First, we focused on the relationship between eWOMs and game consumption, but other factors, such as marketing expenditures and peer influence, may also be relevant. Second, alternative measures of game consumption, such as average playing time, or measures directly linked to game company profit, such as average sales or spending per user, may provide a better understanding of the relationship. Second, there are alternative measures for game consumption, such as the average time a person played. Also, measures that are directly linked to the financial performance of the game companies, such as game sales or average spending for users on each game, may give better insights into establishing a company's strategy. Finally, while this study focuses on online PC games, a significant number of mobile games are also closely linked to social media, such as Facebook game apps. Social network games are strongly associated with social media engagement [31]. While we find that eWOMs are crucial to online games, comparisons between social network games and other

games may provide stronger evidence. We believe that our study can be extended by exploring aforementioned research topics that would be fruitful for future research.

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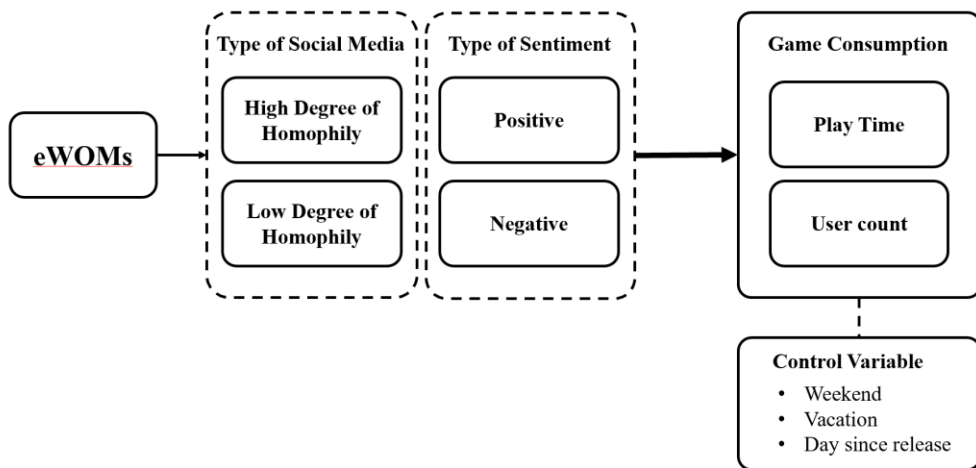


Fig. 1. Modeling Framework.

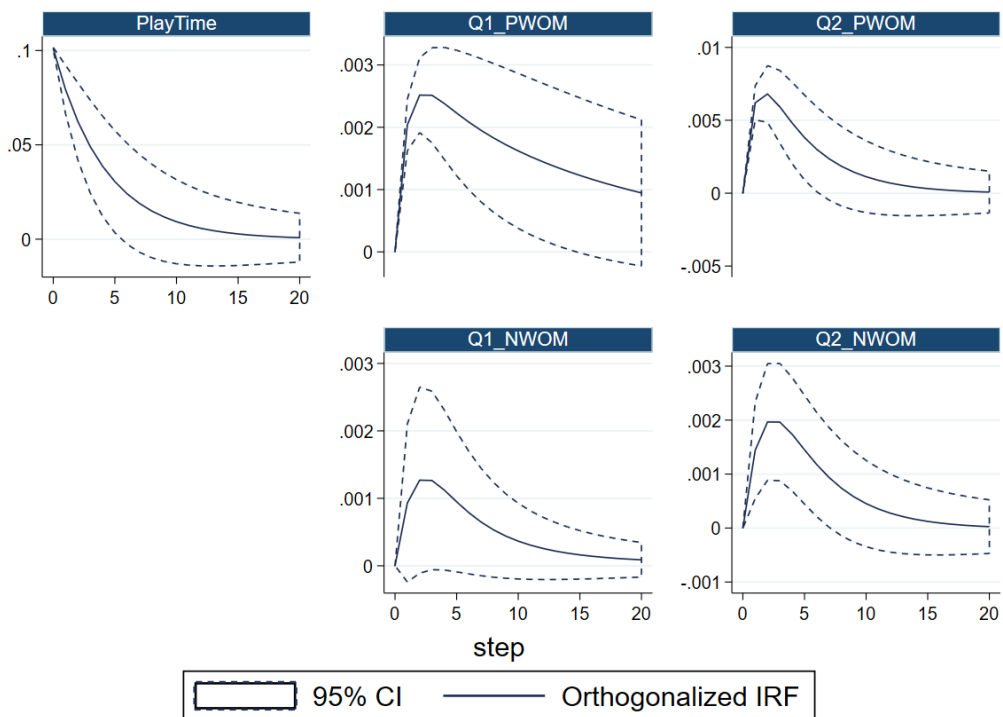


Fig. 2. IRF: response of the daily total sum of hours played for each game brand

Table 1
Social Media Segmentation.

		Algorithm Filtering	
		High	Low
Social Network Homophily	High	Q1: Social media channels belonging to the first quadrant have high communication-oriented characteristics. Users with similar interests tend to group together, and the platform applies algorithmic filtering to posts according to users' interests. <u>Examples:</u> Facebook, Instagram, Kakaostory, Twitter.	Q2: Social media with high algorithmic filtering level and low homophily covers a wide range of topics, and social media selects and presents information to users through algorithms. <u>Examples:</u> Community, New Sites, Forum, Youtube.
	Low	Q3: It provides users with information on a wide range of topics, and gives high accessibility to information exchanged in the media. Users can access desired content without algorithmic filtering, and can communicate with other users in public places. <u>Examples:</u> Blog, Naver Q&A Site	Q4: Users can select the information they want to see (e.g. the follower they want to see) and are exposed only to the selected information. If users select the same information, they will be provided with exactly the same information. Recently, most of the major social media have been filtered, and social media with that characteristic has disappeared. <u>Examples:</u> Twitter before 2016

Table 2

Related Literatures.

Authors	Social Media Platform	Research Findings
Agung & Darma [2]	Instagram	Instagram is operating an algorithmic system based on big data. The algorithmic system Instagram focuses on is cheat prevention and post recommendation. In the post recommendation system, an algorithm is applied in such a way that the user's needs are derived and a suitable post is recommended.
Bakshy et al. [3]	Facebook	Factors such as user's tendency and previous records, along with individual choice limits exposure to ideologically cross-cutting content on Facebook News Feed, while heterogenous friends increase the content diversity.
Choi & Park [14]	Twitter	Twitter network showed the presence of the homophily phenomenon, implying that opinion leaders' influence within the Twitter network was limited to other users sharing the same political views
Cinelli et al. [15]	Twitter, Facebook, Reddit	Homophilic cluster dominate at Facebook and Twitter while Reddit has lower segregation compared to Facebook.
Dubois & Blank [21]	News	When consuming news, high-choice environment allows individuals to consume wide variety of media and the diverse perspective that each media has.
Esteve & Borge [24]	Twitter	Communication in twitter networks is polarized among party and ideological lines with political homophily is prevalent more in retweet network compared to the mention network.
Feezell et al. [25]	News	News based on algorithms made up a large proportion of modern news. Algorithm-based news recommends news that the user is expected to click on based on the user's interests.
Himmelboim et al. [33]	Twitter	Map of the twitter networks of 10 controversial political topics shows that Twitter users are unlikely to be exposed to cross-ideological content from the clusters of users they followed, as these were usually politically homogeneous.
Kitchens et al. [41]	Twitter, Facebook, Reddit	The impacts news consumption has on users differs by platforms. While Facebook and Reddit both increase information source diversity, Facebook shift users toward partisan sites while Reddit shifts toward moderate sites. Increased use of Twitter has little to no change.

Thorson [57]	News	News Recommendation Engine have potential to change patterns of news consumption by allowing readers to communicate both with each other and, indirectly, with news institutions themselves.
Vaidhyanathan [58]	Facebook	Humans tend toward homophily, and Facebook reinforces this as it amplifies human tendency to pursue confirmation and cluster among like-minded individuals.
Vicario et al. [19]	Facebook	Echo chambers spontaneously emerge in Facebook news pages, which are formed by users' interaction. Because of the polarization, groups of users may perceive same topic differently.
Williams et al. [62]	Twitter	Twitter network are characterized by strong attitude-based homophily and segregation into polarized groups. Users are segregated by follower and retweet networks and echo chambers emerge while for the mention network there are interaction among people with mixed-attitudes.
Zamith & Lewis [66]	News	Even if there are some risks of recommendation system bringing problems such as fragmentation, filter bubbles, and homogenization, news commenting forums have possibility for relatively low-cost, easy-to-build systems that could moderate comments more efficiently while also facilitating more civil, cohesive, and diverse discourse.

Table 3
Variable operationalization and summary statistics.

Variables	Definition	Observation	Mean	Std. Dev.	Min	Max
<i>Game Use statistics</i>						
PlayTime	log scaled statistics of the daily total sum of hours played for each game brand	61320	6.4028	0.0035	4.1096	9.7804
UserCount	log scaled statistics of the daily total sum of users who played each game	61320	2.8075	0.0035	0.3010	6.0062
<i>eWOM statistics</i>						
positive eWOM of Q1	postive Volume of eWOM in Facebook, Instagram, and Twitter	61320	9.5207	60.9079	0	1559
negative eWOM of Q1	negative Volume of eWOM in Facebook, Instagram, and Twitter	61320	11.0215	50.9752	0	2582
positive eWOM of Q2	postive Volume of eWOM in News, Community, and Youtube	61320	18.3731	81.6146	0	1460
negative eWOM of Q2	negative Volume of eWOM in News, Community, and Youtube	61320	8.6775	43.0375	0	2056

* The computation is done for individual games each day

Table 4
Coefficient estimates of the proposed panel VAR model.

Independent Variables	Dependent Variable: PlayTime _{i,t}	
	Coefficient	Std. Error
PlayTime _{i,t-1}	0.7855***	0.0641
Q1_PWOM _{i,t-1}	0.0481**	0.0188
Q1_NWOM _{i,t-1}	-0.0828**	0.0381
Q2_PWOM _{i,t-1}	0.1634***	0.0161
Q2_NWOM _{i,t-1}	0.1029***	0.0332
Seasonality	Yes	
Time dummy	Yes	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 5
FEVD result.

Days Ahead	PlayTime _{<i>i,t-1</i>}	Social Media Channels			
		Q1_PWOM _{<i>i,t-1</i>}	Q1_NWOM _{<i>i,t-1</i>}	Q2_PWOM _{<i>i,t-1</i>}	Q2_NWOM _{<i>i,t-1</i>}
1	100.00%	0.00%	0.00%	0.00%	0.00%
2	99.70%	0.03%	0.01%	0.23%	0.01%
3	99.50%	0.05%	0.01%	0.41%	0.03%
4	99.30%	0.07%	0.02%	0.52%	0.04%
5	99.30%	0.09%	0.02%	0.58%	0.05%
6	99.20%	0.11%	0.03%	0.62%	0.06%
7	99.10%	0.12%	0.03%	0.64%	0.06%
8	99.10%	0.14%	0.03%	0.65%	0.07%
9	99.10%	0.15%	0.03%	0.66%	0.07%
10	99.10%	0.16%	0.03%	0.66%	0.07%

Q2_PWOM > Q1_PWOM > Q2_NWOM > Q1_NWOM

Table 6
Coefficient of Subsamples divided by Genre and Social Interaction Features.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Whole Sample	RPG	FPS	RTS	Sports	Action	Arcade	Teamplay Games	Non-Teamplay Games
Dependent Variable: PlayTime _{<i>i,t</i>}									
PlayTime _{<i>i,t-1</i>}	0.7746*** (0.0641)	0.8006*** (0.0816)	0.7908*** (0.1026)	0.8314*** (0.1545)	0.8593*** (0.1595)	0.68 (0.5078)	0.2286 (0.6457)	0.7746*** (0.0641)	0.8791*** (0.2797)
Q1_PWOM _{<i>i,t-1</i>}	0.0524*** (0.0196)	0.0447** (0.0181)	-0.0032 (0.0627)	-0.1097 (0.0853)	0.0779 (0.1229)	0.9689 (2.1866)	0.3129 (0.3248)	0.0524*** (0.0196)	0.3562 (3.6547)
Q1_NWOM _{<i>i,t-1</i>}	-0.0779** (0.0376)	-0.0535 (0.0395)	-0.3492** (0.1771)	-1.4359 (1.0121)	-0.1682 (0.1234)	1.365 (1.3961)	0.8346 (0.8647)	-0.0779** (0.0376)	-0.2001 (0.6491)
Q2_PWOM _{<i>i,t-1</i>}	0.1771*** (0.0147)	0.1470*** (0.0310)	0.2209*** (0.0398)	0.3688*** (0.0926)	0.2782*** (0.1070)	1.1237*** (0.3858)	0.0132 (0.0843)	0.1771*** (0.0147)	0.2124 (0.2216)
Q2_NWOM _{<i>i,t-1</i>}	0.1132*** (0.0329)	0.1201** (0.0567)	0.3360*** (0.1129)	-0.3261** (0.1536)	0.2226* (0.1209)	0.0205 (0.6266)	-0.7187 (0.7354)	0.1132*** (0.0329)	-0.3211 (0.7319)
Seasonality	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Observations	60,984	21,054	7,260	4,356	3,993	5,445	18,876	46,101	15,246

Notes: Standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 7
FEVD result for subsamples divided by Social Interaction Features.

Teamplay Games					
Days Ahead	PlayTime $_{i,t-1}$	Social Media Channels			
		Q1_PWOM $_{i,t-1}$	Q1_NWOM $_{i,t-1}$	Q2_PWOM $_{i,t-1}$	Q2_NWOM $_{i,t-1}$
1	100.00%	0.00%	0.00%	0.00%	0.00%
2	99.59%	0.04%	0.01%	0.34%	0.02%
3	99.25%	0.08%	0.03%	0.60%	0.04%
4	99.04%	0.11%	0.04%	0.75%	0.06%
5	98.90%	0.14%	0.05%	0.84%	0.08%
6	98.82%	0.16%	0.06%	0.88%	0.09%
7	98.76%	0.18%	0.06%	0.91%	0.09%
8	98.72%	0.20%	0.06%	0.93%	0.09%
9	98.70%	0.21%	0.06%	0.93%	0.10%
10	98.67%	0.23%	0.06%	0.94%	0.10%
Average	99.05%	0.13%	0.04%	0.71%	0.07%
Non-Teamplay Games					
	PlayTime $_{i,t-1}$	Social Media Channels			
		Q1_PWOM $_{i,t-1}$	Q1_NWOM $_{i,t-1}$	Q2_PWOM $_{i,t-1}$	Q2_NWOM $_{i,t-1}$
1	100.00%	0.00%	0.00%	0.00%	0.00%
2	99.99%	0.00%	0.00%	0.01%	0.00%
3	99.98%	0.00%	0.01%	0.01%	0.01%
4	99.97%	0.00%	0.01%	0.01%	0.02%
5	99.96%	0.00%	0.01%	0.01%	0.02%
6	99.95%	0.00%	0.01%	0.01%	0.03%
7	99.95%	0.00%	0.01%	0.01%	0.03%
8	99.94%	0.00%	0.01%	0.01%	0.04%
9	99.94%	0.00%	0.01%	0.01%	0.04%
10	99.94%	0.00%	0.01%	0.01%	0.04%
Average	99.96%	0.00%	0.01%	0.01%	0.02%

Table 8
Coefficient estimates of the proposed Panel VAR model.

Independent Variables	Dependent Variable: UserCount _{<i>i,t</i>}	
	Coefficient	Std. Error
UserCount _{<i>i,t-1</i>}	0.8454***	0.0599
Q1_PWOM _{<i>i,t-1</i>}	0.0324*	0.0179
Q1_NWOM _{<i>i,t-1</i>}	-0.1226**	0.0541
Q2_PWOM _{<i>i,t-1</i>}	0.1645***	0.0116
Q2_NWOM _{<i>i,t-1</i>}	0.1185***	0.0427
Seasonality	Yes	
Time dummy	Yes	

*** p<0.01, ** p<0.05, * p<0.1

Table 9
Coefficient of Subsamples with UserCount_{*i,t*} as dependent variable.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	whole sample	RPG	FPS	RTS	Sports	Action	Arcade	Teamplay Games	Non-Teamplay Games
Dependent Variable: UserCount _{<i>i,t</i>}									
UserCount _{<i>i,t-1</i>}	0.8454*** (0.0599)	0.8421*** (0.0894)	0.7869*** (0.0730)	0.8537*** (0.1463)	0.8795*** (0.1491)	0.8226** (0.3530)	0.7886** (0.3236)	0.8305*** (0.0599)	0.9519*** (0.2397)
Q1_PWOM _{<i>i,t-1</i>}	0.0324* (0.0179)	0.0437** (0.0172)	-0.0229 (0.0511)	0.0318 (0.1910)	0.0343 (0.1030)	-0.1593 (1.9501)	0.0843 (0.3126)	0.0367** (0.0184)	0.6208 (4.1890)
Q1_NWOM _{<i>i,t-1</i>}	-0.1226** (0.0541)	-0.0844* (0.0492)	-0.7409** (0.3000)	-3.9478*** (1.3768)	-0.1923 (0.1320)	-0.0677 (2.0699)	-0.2491 (0.2601)	-0.1131** (0.0524)	-0.1208 0.7726
Q2_PWOM _{<i>i,t-1</i>}	0.1645*** (0.0116)	0.1576*** (0.0207)	0.1845*** (0.0377)	0.1765 (0.1764)	0.2421*** (0.0845)	0.9331*** (0.2094)	0.0780*** (0.0178)	0.1737*** (0.0121)	0.2805* (0.1453)
Q2_NWOM _{<i>i,t-1</i>}	0.1185*** (0.0427)	0.1144** (0.0550)	0.5739*** (0.1868)	-0.5522* (0.2905)	0.2441** (0.1122)	-0.2227 (0.6133)	0.2011 (0.2150)	0.1195*** (0.0405)	0.0575 (0.3849)
Seasonality	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time dummy	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
# of Observations	60,984	21,054	7,260	4,356	3,993	5,445	18,876	46,101	15,246

Notes: Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1